Oxidation Numbers and Redox Reactions

Oxidation numbers, agents, half-equations and balancing in acid

Name _____ Class _____ Date _____

RULES AND STEPS

Element: 0 · Monatomic ion: its charge · F: −1 · H: +1 (−1 in metal hydrides) · O: −2 (−1 in peroxides) · Sum = overall charge

Balancing in acid: atoms → O with H_2O → H with H^+ → charge with e^- → equalize electrons → add → check

Check both atoms and charge in every final equation.

A. Understanding

1. Define oxidation and reduction in terms of electrons and of oxidation numbers.

B. Oxidation numbers

2. Give the oxidation number of the named element: (a) N in NH_3 (b) N in HNO_3 (c) Cr in $Cr_2O_7^{2-}$ (d) S in SO_4^{2-} (e) Fe in Fe_2O_3 (f) O in H_2O_2 (g) C in CO_2 (h) Cl in ClO_3^-

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C. Agents and half-equations

3. In $\mathrm{Cl_2 + 2\,Br^- \longrightarrow 2\,Cl^- + Br_2}$, identify the species oxidized, the species reduced, the oxidizing agent and the reducing agent.

4. Write the two half-equations for $\mathrm{Mg + 2\,H^+ \longrightarrow Mg^{2+} + H_2}$ and label each as oxidation or reduction.

5. Which of these are redox reactions? Explain. (a) $\mathrm{NaOH + HCl \longrightarrow NaCl + H_2O}$ (b) $\mathrm{2\,H_2O_2 \longrightarrow 2\,H_2O + O_2}$ (c) $\mathrm{AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3}$

D. Balancing

6. Balance: $\mathrm{Cu + Ag^+ \longrightarrow Cu^{2+} + Ag}$

7. Balance in acidic solution: $\mathrm{Cr_2O_7^{2-} + Fe^{2+} \longrightarrow Cr^{3+} + Fe^{3+}}$

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8. Balance in acidic solution: $\mathrm{Cu} + \mathrm{NO_3^-} \longrightarrow \mathrm{Cu^{2+}} + \mathrm{NO}$

E. Challenge

9. (a) Balance in acidic solution: $\mathrm{MnO_4^-} + \mathrm{C_2O_4^{2-}} \longrightarrow \mathrm{Mn^{2+}} + \mathrm{CO_2}$. (b) In a titration, 25.00 mL of an $\mathrm{Fe^{2+}}$ solution needs 20.00 mL of 0.0200 M $\mathrm{KMnO_4}$ to react completely ($\mathrm{MnO_4^-} + 8\,\mathrm{H^+} + 5\,\mathrm{Fe^{2+}} \longrightarrow \mathrm{Mn^{2+}} + 4\,\mathrm{H_2O} + 5\,\mathrm{Fe^{3+}}$). Calculate the concentration of $\mathrm{Fe^{2+}}$.

10. Balance in basic solution: $\mathrm{Cr(OH)_3} + \mathrm{ClO^-} \longrightarrow \mathrm{CrO_4^{2-}} + \mathrm{Cl^-}$

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